

HAND IN HAND WITH GPS

The global positioning system is a constellation of 27 satellites, which can pinpoint your location anywhere on earth



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The global positioning system (GPS) is a satellite-based global navigation satellite system (GNSS) that provides various types of information including geographic location, latitude, longitude, altitude, time, etc. It is a reliable system, which is unaffected by weather and time. Anyone can freely access it with a GPS receiver.

GPS receiver

GPS receiver is a device that receives GPS signals for determining the de-

vice's current location on earth. The communication method of the receiver is serial. The serial data can be read through receive and transmit buffers by any embedded system or computer via RS-232 or USB. Mostly Garmin GPS receivers are available in markets and these use NMEA protocols.

NMEA protocol

It is a protocol that defines the transmission of data in the form of sentences from one source to unlimited number of listeners. The GPS receiver receives the information from satel-

lites using NMEA. Important protocol rules are:

1. Message starts with a dollar (\$) sign.
2. Information about message and source is carried by next five characters.
3. Data fields are separated by comma.
4. Null bytes mention unavailable data.
5. '<CR><LF>' at the end of the message.

Important message types of NMEA include:

Global Positioning System Fix Data (GGA). Time, position and fix related data for a GPS receiver.

Geographic Position—Latitude/Longitude (GLL). Latitude and longitude, UTC time of fix and status.

Satellites in view (GSV). Number of satellites in view, satellite ID (PRN) numbers, elevation, azimuth and SNR value.

Recommended Minimum Specific GNSS Data (RMC). Time, date, position, course and speed data.



Fig. 1: GPS receiver